

Power Budget Example

Team Number:	101
Project Name:	Autonomous Irrigation System
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Version:	1

A. List ALL major components (active devices, integrated circuits, etc.) except for power sources, voltage regulators,

All Major Components	Component Name	Part Number	Voltage Range	#	Maximum Current	Current (mA)	Unit
	9V DC Motor	1528-1150-ND	3-9V	1	250	250	mA
	FAN8100N	2156-FAN8100	1.8-9	1	3000	3000	mA
	PIC Curiosity Nano Microcon	DM164142 -PIC1	4.4-5.25	1	0.1	0.1	mA

B. Assign each major component above to ONE power rail below. Try to minimize the number of different power rails in

9V Power Rail		Component Name	Part Number	Voltage Range	Quantity	Maximum Current	Current (mA)	Unit
		9V DC Motor	1528-1150-ND	3-9V	1	250	250	mA
		FAN8100N	2156-FAN8100	1.8-9	1	250	250	mA
							0	mA
							0	mA
							0	mA
							500	mA
							25%	
							625	mA
c1. Regulator or Source Ch		9V		9V	1	1500	1500	mA
							875	mA
+5V Power Rail		Component Name	Part Number	Voltage Range	Quantity	Maximum Current	Current (mA)	Unit
		PIC Curiosity Nano Microcon	DM164142-PIC1	4.4-5.25	1	0.1	0.1	mA
							0	mA
							0	mA
							0	mA
							0	mA
							0.1	mA
							25%	
							0.125	mA
c2. Regulator or Source Ch		+5V Regulator	LM7805	5V	1	1500	1500	mA
							1499.875	mA

[illegible]

C. For each power rail above, select a specific voltage regulator using the same process as for major component							
D. Select a specific external power source (wall supply or battery) for your system, and confirm that it can supply all of the							
External Power Source 1	Component Name	Part Number	Input Voltage	Rated Output Voltage	Maximum Current	Current (mA)	Unit
Power Source 1 Selection	Plug-in Wall Supply	BestCH	240	9V	1.5A	1500	mA
Power Rails Connected to External Power Source 1	+5V Regulator	LM7805	35V	5V	1.5A	1500	mA
Total Remaining Current Available on External Power Source 1						0	mA

Notes

External Supply Voltage should be determined by the dropout voltage for highest-voltage regulator (e.g., +14V for a +12V).
If you have multiple units in your design (e.g., a base unit and remote unit) then you need a separate power budget for each unit.